

AHMEDABAD, India

WMO: 426470

Lat: 23.0766N

Lon: 72.6276E

Elev: 189

StdP: 14.60

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.6	53.7	27.4	21.4	76.5	31.8	26.4	76.2	14.9	72.8	13.5	72.7	2.8	50	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.1	109.6	73.4	107.5	73.2	104.9	73.1	83.4	92.8	82.4	91.1	81.6	90.2	7.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	162.4	86.5	80.7	160.3	86.0	79.7	154.9	85.3	47.8	92.6	46.7	91.9	45.8	91.0	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	13.2	11.8	DB	46.6	112.7	3.3	3.1	44.3	114.9	42.4	116.7	40.5	118.5	38.1	120.7	(4)
(5)			WB	41.1	86.4	2.7	2.2	39.1	88.0	37.5	89.3	36.0	90.6	34.0	92.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	68.6	73.7	82.7	90.3	94.5	91.9	85.8	84.0	85.0	84.1	77.4	70.9	(6)	
	DBStd	8.75	3.82	5.03	4.70	3.72	3.10	4.52	3.72	2.66	2.86	3.38	3.67	3.79	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	18	11	4	0	0	0	0	0	0	0	0	0	3	(9)	
	CDD50	11844	577	664	1014	1208	1380	1257	1109	1054	1051	1058	823	649	(10)	
	CDD65	6387	123	249	549	758	915	807	644	589	601	593	373	186	(11)	
	CDH74	84766	1280	2739	7206	11452	14356	12048	8166	6931	7461	7348	3919	1860	(12)	
	CDH80	47725	335	1232	4260	7475	9919	7776	3867	2755	3453	4067	1958	628	(13)	
(14) Wind	WSAvg	5.2	4.2	4.2	4.4	5.4	7.5	8.2	7.1	6.2	4.8	3.0	3.1	3.8	(14)	
(15) Precipitation	PrecAvg	30.8	0.1	0.0	0.0	0.1	0.4	3.5	11.5	9.6	4.4	0.3	0.3	0.1	(15)	
	PrecMax	52.3	0.6	0.5	1.0	0.9	9.1	11.9	32.3	24.1	19.1	3.8	3.9	1.1	(16)	
	PrecMin	10.4	0.0	0.0	0.0	0.0	0.0	0.0	1.4	0.7	0.0	0.0	0.0	0.0	(17)	
	PrecStd	11.7	0.1	0.1	0.2	0.2	1.3	3.0	6.3	5.8	4.6	0.7	0.8	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	97.0	107.2	111.1	113.4	111.5	102.2	96.4	100.4	102.3	96.8	91.5	(19)	
		MCWB	66.7	67.2	69.5	71.4	74.4	77.3	79.8	81.1	79.3	73.7	70.1	66.1	(20)	
	2%	DB	86.1	93.5	102.8	107.8	111.0	107.5	98.5	93.3	96.9	100.0	94.6	88.1	(21)	
		MCWB	64.2	65.9	67.9	70.4	73.7	77.8	79.8	80.3	79.2	73.8	69.3	64.7	(22)	
	5%	DB	84.0	91.0	100.0	105.7	108.1	104.1	95.0	91.4	94.6	97.7	91.8	86.1	(23)	
		MCWB	63.0	65.4	67.0	70.0	73.7	78.3	79.6	79.7	79.4	73.0	68.4	64.3	(24)	
	10%	DB	81.0	87.7	96.8	102.7	105.9	100.8	91.8	89.5	91.7	95.2	89.9	84.0	(25)	
		MCWB	61.8	64.3	66.8	69.7	73.5	78.6	79.8	79.3	78.8	72.2	67.7	63.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	71.7	74.7	79.6	83.2	84.6	83.9	83.7	85.3	82.8	78.5	70.5	(27)	
		MCDB	83.2	87.9	94.0	91.7	96.0	96.0	92.3	90.8	93.5	92.0	89.8	83.2	(28)	
	2%	WB	67.6	69.2	71.9	77.3	81.5	82.9	82.6	82.2	83.1	80.6	75.3	68.2	(29)	
		MCDB	80.8	87.2	92.4	91.4	94.7	95.4	89.8	87.9	90.6	90.1	84.6	82.9	(30)	
	5%	WB	65.6	67.4	70.1	75.7	80.3	81.9	81.8	81.3	81.7	79.1	72.8	66.5	(31)	
		MCDB	78.9	85.1	91.2	91.3	93.9	94.3	88.7	87.1	88.5	88.0	85.4	81.3	(32)	
	10%	WB	63.7	65.8	68.5	74.2	79.2	81.0	81.0	80.5	80.6	77.8	70.3	64.9	(33)	
		MCDB	77.4	83.6	90.4	92.2	93.3	93.1	87.8	86.3	87.4	86.9	84.8	80.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	27.2	27.4	26.2	24.1	17.7	11.2	10.4	13.7	23.0	26.5	26.4	(35)	
		MCDBR	28.5	29.8	31.0	28.7	26.8	21.8	15.5	13.1	17.7	25.7	27.5	28.7	(36)	
	5% WB	MCWBR	12.2	10.8	10.8	9.5	9.0	6.8	5.3	5.5	7.3	8.4	10.4	11.8	(37)	
		MCWBR	24.7	26.5	26.3	25.3	23.4	17.8	12.0	11.0	13.8	17.7	23.5	25.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.422	0.428	0.484	0.539	0.579	0.689	0.719	0.656	0.591	0.531	0.522	0.462	(40)	
		taud	2.123	2.122	1.977	1.889	1.849	1.668	1.629	1.754	1.878	1.940	1.908	2.011	(41)	
	Ebn at Noon	Ebn at Noon	263	271	261	248	237	211	205	218	230	237	229	243	(42)	
		Edh at Noon	45	48	58	64	67	79	82	73	63	57	56	49	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1433	1725	2052	2287	2342	1895	1319	1312	1603	1743	1465	1344	(44)	
	RadStd	RadStd	46	39	57	46	73	131	114	125	135	75	78	64	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.86	N/A	N/A	+1.58	N/A	N/A	N/A	N/A	+311	+310
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page